Skin Lesion Segmentation using Modified SegNet Model

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This is to certify that the work presented in the thesis entitled **"Skin Lesion Segmentation using Modified SegNet Model"** in partial fulfillment of the requirement for the award of degree of **Bachelor of Computer Application** of Institute of Engineering & Management is an authentic work carried out under my supervision and guidance.

To the best of my knowledge the content of this thesis does not form a basis for the award of any previous Degree to anyone else.

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CERTIFICATE OF APPROVAL

The foregoing thesis entitled **"Skin Lesion Segmentation using Modified SegNet Model"** is hereby approved as a creditable study of research topic and has been presented in satisfactory manner to warrant its acceptance as prerequisite to the degree for which it has been submitted.

It is understood that by this approval, the undersigned do not necessarily endorse any conclusion drawn or opinion expressed therein but approve the thesis for the purpose for which it is submitted.

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Abstract

Analysis of dermoscopic images has become an important field of research. Using computers to do this work can help in eliminating observer bias. Dermoscopic images are produced by dermoscope, a special purpose dermatology instrument. They usually have uniform illumination and have more contrast. One of the most important steps in performing this analysis is to detect the lesion areas and segment them automatically. This would help in diagnosing any kind of disease at a much faster rate. Segmenting the lesion site from healthy skin is a crucial step in skin disease diagnosis.

In this study, a deep learning-based method for precisely extracting the lesion region is proposed. A deep convolutional neural network is used to process the input images, after performing some preprocessing on it. We use PH2 2013 dataset, which contains dermoscopic images along with their corresponding lesion binary masks. These lesion binary masks will be the ground truth which is used to evaluate the results. The SegNet architecture used in our study achieved an accuracy of 99.14, with benchmark standard of DC 96.94 and JI 99.11.

The progress in this field is encouraging and can lead to fully-fledged automated approaches for skin lesion segmentation. We hope to contribute to its progress.

Chapter 1

1.1 Introduction

We provide a deep CNN architecture in this research to extract the lesion region from skin images. Skin lesion segmentation is a crucial step in the skin diagnostics process because it enhances both manual and computer-aided diagnostics by directing medical staff attention to certain areas of the skin. Deep neural networks are effective for the typical computer vision problem of image segmentation, which divides a digital image into many parts. For diagnosing patients, tracking treatment progress, and evaluating clinical trials for novel therapies, automatic and precise segmentation of lesions is crucial. From the detection of skin lesions to the identification of brain lesions, lesion segmentation is essential for many activities related to medical diagnosis.

Commonly used segmentation techniques include image analysis methods, such as the threshold method [1]–[3], active contour model [4], clustering method[5], [6], histogram statistical method [7] and region-growing method [8]–[10], and supervised learning methods such as support vector machine (SVM) [11] and deep learning. These methods use the object's own data (such as colour and texture [12]) or learning from a labelled training dataset to perform the segmentation. For reviews of pigmented skin lesion segmentation methods, refer to [13]

In recent years, with the development of parallel processing capabilities using graphical processing units (GPUs) and the availability of large training datasets (such as ImageNet [14]), classification and recognition methods based on deep convolutional neural networks (CNNs) have attracted considerable attention. From the early LeNet [15] to the later AlexNet [16] and VGG-16 [17].

This project's primary purpose is to recognize lesions in a dermoscopic picture and separate the lesions from the surrounding skin.

Chapter 2

2.1 Background Studies

Dermoscopy is a non-invasive skin imaging technique that is used for the diagnosis of skin lesions. Using these dermoscopic skin lesion images, doctors can visualise the sub-surface structures more effectively. However, even with dermoscopy, diagnostic accuracy ranges from 24% to 77% depending on the clinician's level of expertise. Inexperienced dermatologists may even lower the accuracy of diagnosis when using this method. Therefore, to minimize the diagnostic errors that result from the difficulty and subjectivity of visual interpretation, and limited access of these instruments to dermatologists, the development of Computer Aided Diagnosis (CAD) systems is crucial.

The main hindrance in assessing these images is caused by simple characteristic changes that vary from patient to patient, such as the presence of hair, varying sizes, colour, and shapes of moles. Even growth aspects such as asymmetrical lesions with irregular boundaries can be of importance. Or even due to the poor contrast between the normal skin and lesion.

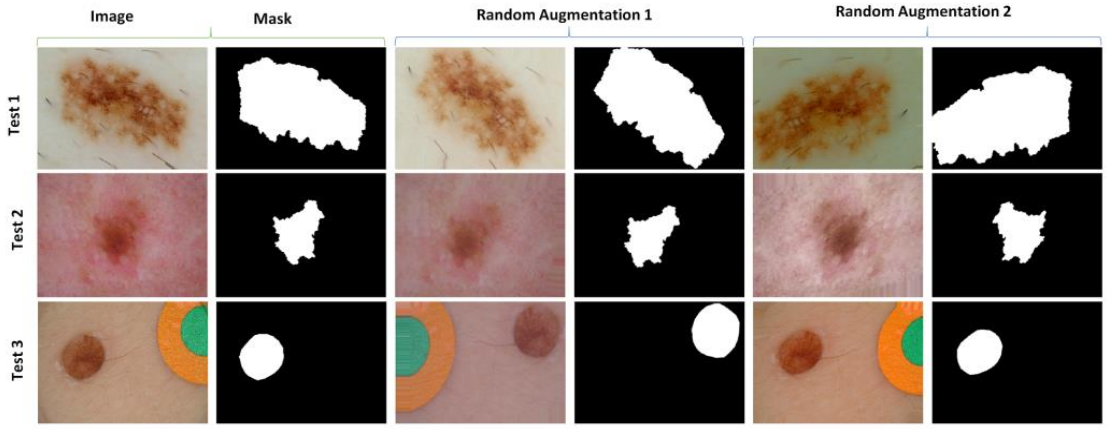
2.2 Literature Survey

In the study [18], they propose a CNN-based skin lesion segmentation framework, called SkinNet. The proposed CNN architecture is a modified version of the U-Net.

Melanoma, the foremost perilous shape of skin cancer, was anticipated to have killed near to 60,000 individuals and harrowed over 350,000 individuals universally in 2015. Despite the high mortality rate, early location increments survival to almost 95%. As of late, profound learning strategies based on convolutional neural semantic segmentation. In this research, compared to

hand-crafted features, CNN-based methods can extract more noticeable features from the entire image.[18]

This paper provides the SkinNet, a framework for segmenting skin lesions based on CNN. They have updated the U-Net version to create this CNN architecture.
The essential components of the U-net architecture are an encoder (a contracting path) that down samples an image into a set of high-level features and a decoder (a symmetric expanding path) that builds a pixel-wise segmentation mask using the feature information. To give the features extracted from the image, a more comprehensive context, SkinNet uses dilated convolutions in the lowest layer of the encoder branch in the U-Net.
To incorporate multi-scale visual information more effectively, dense convolution blocks are used in place of the traditional convolution layers in both the encoder and decoder branches of U-Net. They have two convolutions plus ReLu activation layers in each dense block in both the encoder and decoder sections. Also, the convolution layers that follow the output and the input feature map are combined.



Dermoscopy images augmentation. The first and second columns show the training images with their masks and columns 3-6 shows augmented images with various rotation, flipping, color shifting, translation and scaling operations.

Data Augmentation in paper [18]

All training photos were first adjusted using the min-max normalization algorithm and then downsized to 512 X 512 pixels because most of the images

in the training dataset varied in look and size. Additionally, they enhanced the data to improve SkinNet's performance. Various picture modifications, including rotation, flipping, colour shifting, translation, and scaling procedures, are used to randomly enrich the input images.

They have trained SkinNet using the ISBI 2017 challenge dataset, which has 2000 dermoscopic pictures and the matching lesion masks, to assess its performance. By comparing the network's lesion segmentation accuracy to the given ground truth masks and calculated the Dice similarity coefficient (DC) and the Jaccard index, two well-known similarity measures. To directly compare their technique with the state-of-the-art, as given in recent research, they have additionally estimated the sensitivity (SE), specificity (SP), and accuracy (AC) metrics. Comparing their method to other innovative techniques that made use of different CNN architectures, they were able to attain greater JI and DC.

Machine learning methods started gaining more recognition for segmentation tasks when clustering, pattern recognition techniques and supervised classification gave exceptional results in skin lesion feature extraction for segmentation. A general CNN architecture for image segmentation consists of a series of layers of convolution filters, interspersed with pooling layers. Then these are flattened to a vector for input to the fully connected layers.[19]

The development of a method for automatic analysis of dermoscopy images is necessary to improve diagnosis and reduce death. Automatically segmenting melanoma from the surrounding skin is a crucial step in automatic analysis of melanoma. However, this is not easy, as melanoma lesion formation can vary in size, shape, and colour, for diverse types of skin and texture.
The rate of growth of the lesion, the shape and its irregularities, and its spreading are all compared to every person and even unique in some cases. A machine can sometimes pick up features which may not be noticed by a human or may not be even visible to the naked eye. It can be crucial in effectively treating the patient.

Lesion Segmentation output in paper [19]

Deep learning methods have shown favourable outputs in various pattern identification and medical imaging applications. Convolutional neural networks (CNN) are widely used deep learning methods, which are inspired by the human visual cortex. A CNN is used to extract vessels in fundus images and for brain tumour segmentation. The guided filter is used as an edge-preserving smoothing operator. It reduces the effect of noisy factors while causing minimal distortion on the lesion's border. Noisy effects and irrelevant textures, that could misdirect the segmentation procedure, are smoothed out. A crucial step in computerized analysis of melanoma is to locate the exact region of pigmented lesion in a skin image.[20]

This means that the image must be segmented into two regions as lesion and normal skin. Performances of other steps that follow, such as feature extraction and classification, are directly depended on the accuracy of the segmentation step. Quality of segmentation can highly affect the assessment of metrics such as border irregularity and lesion asymmetry. There are many algorithms for automatic extraction of lesion region in skin images. These methods can be categorized in three groups of thresholding, active contours, and region merging methods.

Internet of Medical Things (IoMT) is a term that describes an application and inclusion of medical devices which link technology in the field of healthcare with the application of networking technologies, models. This decreases the number of unneeded hospital visits. The cost to healthcare systems of connecting patients to enables them to speak with their respective doctors the health-related information via a secure network.[21]

On the other hand, skin lesions can be cancerous that can widely spread and affects people all over the world. Several types of skin cancer, including intraepithelial carcinoma, basal cell carcinoma (BCC), and melanoma.

For automated identification of skin lesion images, a set of four main processes exist namely pre-processing, segmentation, feature extraction, and classification. To find the presence of cancer that might be caused through the skin lesions, lesion segmentation is found to be basic process. But it becomes difficult because of the major dissimilarity in location and size of the skin lesion images. In addition, the poor contrast of the image put off the discrimination of the neighbouring tissue cells. Additionally, further constraints like air bubbles, hair, ebony frame, ruler mark, blood vessel, and colour illumination pose even more difficulty for segmenting skin lesion images.

The presence of essential features present in the dermoscopic images acts a vital part in the melanoma classification. They can be attained through effectively segmenting the lesion from the adjacent tissue cells. The process of segmenting lesion from the neighbouring healthy tissue and filtering notable features are important for robust and effectual identification process.[21]

Deep learning (DL) models particularly convolution neural network (CNN) received more attention in different areas of image classification, segmentation, object identification and so on. It is effective due to the significant learning process of hierarchical features and filters maximum robust features from new image. Various stages of CNN are employed in different processes such as classification, localisation, and segmentation.

In the paper[22], the crucial skin lesion has been segmented, with semantic segmentation the objects within an image are segmented and classified based on a concept. It can be performed in 2D images, videos or even 3D data and is still a challenging task. Recently, convolutional neural networks (CNNs) have achieved remarkable results in computer vision tasks. Segmentation based on clustering most commonly uses the K-means algorithm to gather samples into different clusters according to the distance.
The researchers used one of the most efficient threshold segmentation algorithms is the largest inter-class variance method (Otsu) VGG16, a CNN composed of 16 layers and achieved 92.7% accuracy on ILSVCR Caltech-101 dataset. Two approaches for image segmentation of breast cancer images have been presented. They are the first group of researchers to develop a machine learning model that can analyse 3D MRI images of brain lesions. This model has been trained end to end on MRI volumes being prostate, pro- posed and has

shown good performances on demanding datasets. This model using Dice-based loss has received a score of 82.39, which are comparable to other researchers who were the winners of that competition.

The models for melanoma detection proposed in this paper have been evaluated on the ISIC 2018 competition. Several pre-processing techniques were needed to face the challenge of having images of varying size and type. They used several types of transformations of the original images, such as rotation, translation, flipping, etc., resulting in an increased number of images corresponding to the training iterations.

The evaluated approaches are promising and could be used in practical applications, especially as more labelled images with skin lesions are collected.

The automated skin lesion analysis workflow's tough and crucial operation is the process of segmentation. Size, border definition, symmetry and irregularity of shapes are crucial factors in the diagnosis of skin cancer.

Localization and delineation of lesions are necessary for both surgical excision and radiation therapy (American Cancer Society, 2020). The tedious process of manual delineation is hampered by high inter- and intra-observer variability. Therefore, an efficient and precise segmentation algorithm is crucial to a successful computer-aided diagnosis (CAD) system for skin cancer. Recent studies prove the significance of segmentation in enhancing the performance of classification for several diagnostic categories by enabling the cropping of lesion images and the eradication of imaging artifacts.[23]

Chapter 3

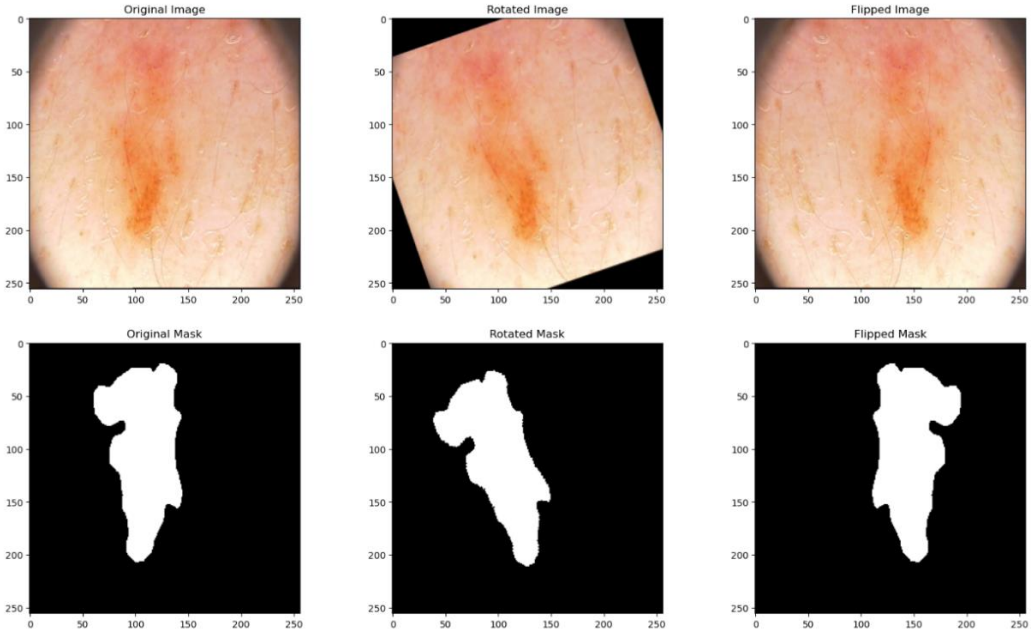
3.1 Proposed Methodology

I. Pre-processing

The PH2 dataset consisting of 200 dermoscopic images is used. The dataset is split into validation and test set in the ratio 4:1. Thus resulting in a training set of 180 images and test set of 20 images. The images are resized into 256 X 256-pixel images.

Image augmentation is performed on the training set such as horizontal flip and random rotation. The training dataset increases to 540 images. This larger dataset is split into training and validation set which is fed to the neural network.

Image Augmentation examples:



II. Convolutional Neural Network

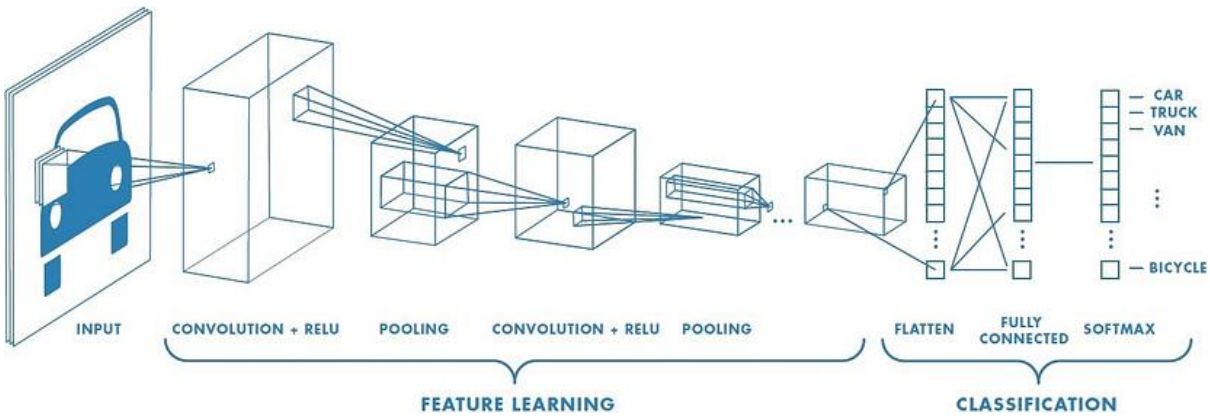


Fig. 1. An illustration of CNN architecture [29].

The first step is to make a Convolutional layer in our network with matrices of Feature Detectors. We feed this layer with standardized sizes of images with the activation function Rectified Linear Unit (ReLu). We apply Max Pooling to further

reduce space invariance, reduce the size of our image and still preserve the important features detected by the Feature Detectors. We then Flatten the reduced feature map to convert this matrix into a column of vector. These will be the input to our hidden layer(s). These hidden layers will have several neurons which also have the ReLu activation function, with a layer to give an output with a sigmoid function. Our neural network is now ready.

The next step is to feed the images into this neural network. By doing image augmentation, we can have a smaller dataset converted into a larger one for our CNN to learn from. This real time augmented dataset is fed into CNN and CNN learns from it in epochs. After each epoch it checks for its loss function, which is used to measure its accuracy. It then readjusts its weights and continues to learn. A test set is also present to validate its predictions and measure its accuracy.

III. Image Segmentation

There is a transition from ways of image processing where the inputs and outputs are both pictures to methods where the inputs are still images. Still, the outputs are characteristics taken from those images.

We use the SegNet semantic segmentation model which is an architecture made up of encoders and decoders. A pixel wise classification layer follows it.

A picture is segmented into its individual areas or objects. The problem to be solved determines how far the subdivision is taken. When an application's objects or regions of interest are identified, segmentation should come to an end. Since one of the most challenging jobs in image processing is segmenting non-trivial pictures. One or two fundamental characteristics of intensity values, discontinuity, and similarity are the foundation of segmentation algorithms. Our approach is based on partitioning an image into regions that are similar according to a set of predefined criteria. like, regions developing, combining, and dividing. The picture is divided or partitioned into several fragments. We may use the crucial segments from the image after segmenting it to process the image.

IV. Architecture

SegNet is a type of deep learning algorithm that is used for a specific type of image processing called semantic segmentation. Semantic segmentation involves identifying and classifying objects within an image at the pixel level. There are no fully connected layers and hence it is only convolutional. It is made up of two parts: an "encoder" network and a "decoder" network. A decoder upsamples its input using the transferred pool indices from its encoder to produce a sparse feature map(s). It then performs convolution with a trainable filter bank to densify the feature map. The final decoder output feature maps are fed to a soft-max classifier for pixel-wise classification. An advantage is that it is specifically designed for analysing images and identifying objects at the individual pixel level. This makes it good at identifying minute details in images that other types of models might miss.

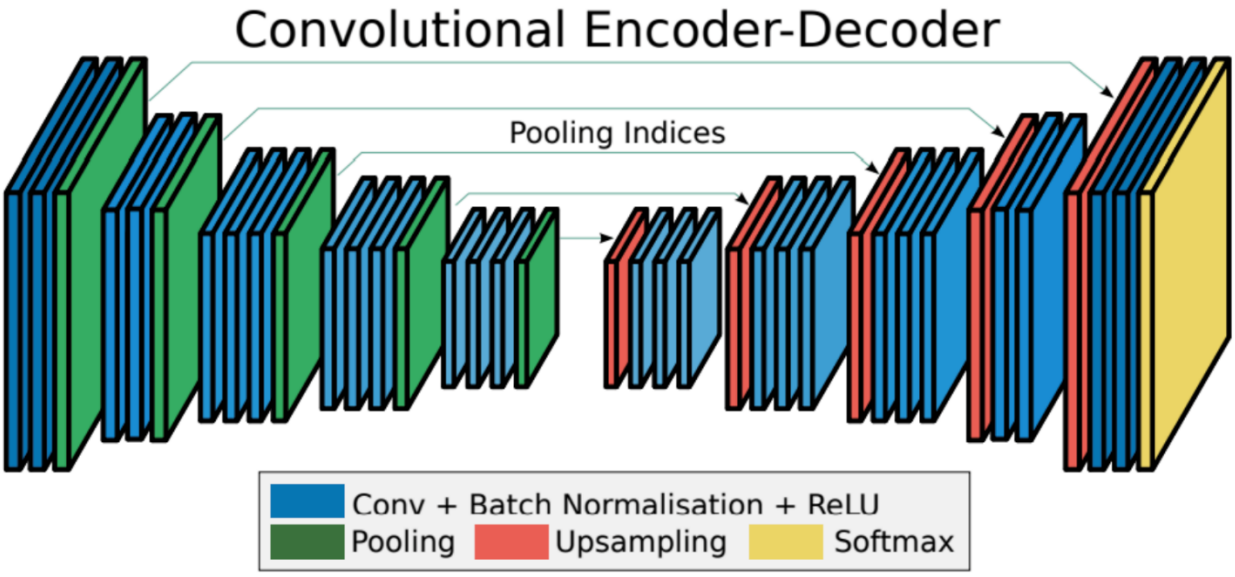
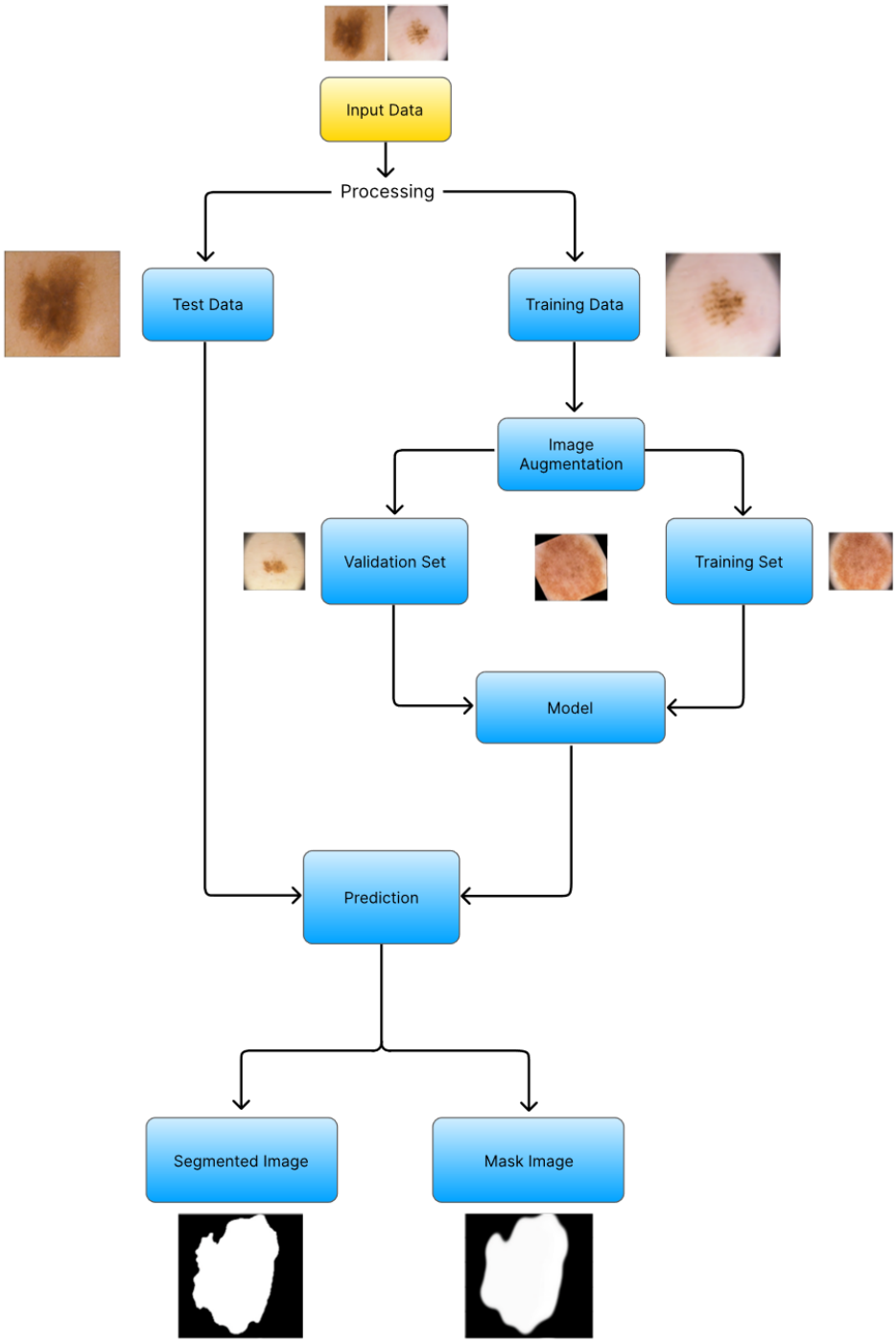


Fig. 2. An illustration of the SegNet architecture [24].

SegNet has not yet been used in research work in the areas of medical image segmentation. Since it is a powerful architecture, it promises reliable results. Skin lesion segmentation has a huge scope for automation and this architecture can take the work forward. Hence, we have used this as the basis of our research in this area.

V. Flowchart:


Input Data
Processing
Test Data
Training Data
Image Augmentation
Validation Set
Training Set
Model
Prediction
Segmented Image
Mask Image

The first step is to do image augmentation on our training set to increase the diversity and number of images in our training set. We then split this augmented dataset into a validation set and the training set. We feed this into our model. The model gives us the prediction. The output of this prediction is the mask of the image and the segmented lesion from the original image. The model tests its accuracy on the test set based on the evaluation metrics that we use.

Chapter 4

4.1 Experimental Dataset

IDE	Anaconda, Jupyter Notebook
Platform	Python
Libraries	TensorFlow keras, OpenCV, NumPy, panda, PIL, glob, matplotlib

Dataset:

Link: https://www.fc.up.pt/addi/ph2%20database.html

Total Images: 200 images + Corresponding ground truth

Split into:

Training Set: 180 images + Corresponding ground truth

Test Set: 20 images + Corresponding ground truth

Data Augmentation: The dataset is subjected to Data Augmentation. The size of the augmented dataset increases from 180 to 540 for training out model.

Chapter 5

5.1 Results and Discussions

The prediction of our model will result in the outputs of two forms, mask of the image and the segmented image. The resultant image will show the output as the original image, the ground truth, and its predicted image, which will be generated by our model after being trained by the training and validation sets. We made use of a validation set to test the data while it was being trained. Along with the images we propose to display the classifiers of our model which will

differentiate it from other papers. With the use of Jaccard Index or Intersection Over Union (IoU), Dice Coefficient (DC), Precision and Accuracy which be proven to be a good measure to find the efficiency of our model.

Accuracy: The accuracy is defined as the ratio of the number of correct predictions to the total number of predictions made by the model.

$$\textbf{Accuracy} = \frac{\boldsymbol{True_{positive}} + \boldsymbol{True_{negative}}}{\boldsymbol{True_{positive}} + \boldsymbol{True_{negative}} + \boldsymbol{False_{positive}} + \boldsymbol{False_{negative}}}$$

Dice Coefficient: The Dice coefficient is a commonly used metric for measuring the similarity between two sets of data, such as two binary images or two sets of segmented regions. It is defined as the ratio of the intersection of the two sets to their average size.

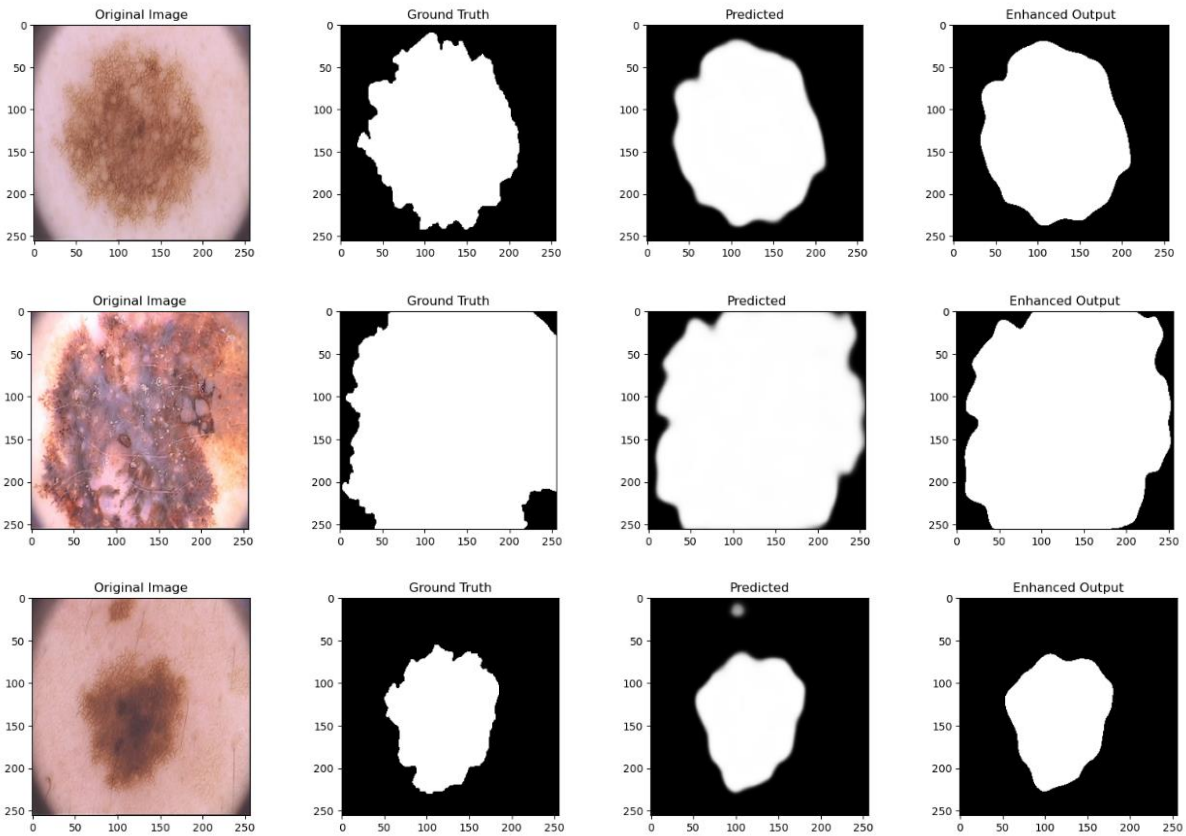
$$\textbf{Dice} = \frac{\boldsymbol{2\ X\ (True_{positive})}}{\boldsymbol{(True_{positive}+False_{positive})+(True_{positive}+False_{negative})}}$$

Jaccard Index (Intersection over Union): Jaccard index is be used to evaluate the similarity between two segmentation masks, one being the ground truth and the other being the predicted segmentation.

Jaccard Index = $\frac{Intersection}{Union}$

We achieved an accuracy of 99.14, with benchmark standard of DC 96.94 and JI 99.11.

Segmentation Results:



A comparison of our results with other papers

Ref	Year	Dataset	DC	JI	ACC
Peng Tang et al.[25]	2019	ISIC 2017	86.93	79.26	94.31
Xie et al.[26]	2019	PH2	91.9	85.7	94.9
DCL-PS[27]	2019	PH2	92.10	85.90	95.30
FRCN[28]	2018	PH2	91.77	84.79	95.08
Peng Tang et al.[25]	2019	PH2	94.23	94.23	97.60
Our Work	**2023**	**PH2**	**96.94**	**99.11**	**99.14**

Table 1. Results of Deep learning models for Skin Lesion Segmentation

Chapter 6

6.1 Conclusion

In this paper, we presented a novel approach for skin lesion segmentation using convolutional neural networks. We used the PH2 dataset for training and evaluation of our model and employed the SegNet architecture for image segmentation. Our results show that our model achieved an accuracy of 99.15% in segmenting skin lesions from medical images, which is a remarkable improvement over existing segmentation technique.

The success of our approach can be attributed to the ability of the SegNet architecture to incorporate both local and global information, allowing it to capture minute details in the images while also keeping spatial context. Furthermore, our use of data augmentation techniques such as flipping and

rotation during training helped to improve the generalization capability of the model and reduce overfitting.

Our findings have significant implications for medical image analysis, particularly in the field of dermatology. The ability to accurately segment skin lesions can aid in the diagnosis and treatment of skin diseases, potentially leading to improved patient outcomes. Future work could explore the use of our approach on larger datasets and investigate the transferability of our model to other medical image analysis tasks.

In conclusion, our study proves the effectiveness of using convolutional neural networks and the SegNet architecture for skin lesion segmentation, achieving an accuracy of 99.15% on the PH2 dataset. Our results suggest that this approach could have important applications in medical image analysis and pave the way for further research in this area.

6.2 Future Work

To further improve our architecture and model to produce better results and better efficiency in this model.

We will use Jaccard Index, Dice Coefficient, Accuracy, Sensitivity, Specificity, and such other benchmarks to evaluate the performance of our Convolutional Neural Networks and compare them with the ground truth of our dataset.

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